📊 Statistics

📃 Dataset

Name	Gender	Marks
Asha	F	78
Ravi	F	85
Meena	F	78
Kiran	M	92
Sita	F	67
Arun	M	85
Divya	F	74
Manoj	F	90
Pooja	F	78
Ramesh	F	60

📌 Basic Statistical Measures

1️⃣ Mean (Average Score)

Mean is the average of all marks.

Formula:

Mean = (Sum of all marks) / (Number of students)

Calculation:

= (78 + 85 + 78 + 92 + 67 + 85 + 74 + 90 + 78 + 60) / 10
= 787 / 10
= **78.7**

👉 **Average Score = 78.7**

2️⃣ Median (Middle Value)

Step 1: Arrange marks in ascending order

60, 67, 74, 78, 78, 78, 85, 85, 90, 92

Step 2: Since total students = 10 (even number)

Median = Average of 5th and 6th values
= (78 + 78) / 2
= **78**

👉 **Median Value = 78**

3️⃣ Mode (Majority Category – Gender)

Mode is the most frequently occurring value.assume:

Gender count:

- F (Female) → 8 students
- M (Male) → 2 students

👉 **Mode = female**

📊 Average Height of Students

Step 1: Question

In an organization, we want to find:

👉 **What is the average height of students?**

📊 Step 2: Dataset (Sample Data)

Student Name	Height (cm)
Asha	150
Ravi	160
Meena	155
Kiran	170
Pooja	165
Arun	158
Divya	162
Rahul	168
Sneha	159
Mohan	161

📘 Step 3: Definition of Statistics

Statistics is the process of:

- Collecting data
- Organizing data
- Analyzing data
- Interpreting data

👉 In simple words:
Statistics helps us understand data and make decisions.

📗 Step 4: Types of Statistics

1️⃣ Descriptive Statistics

Definition:
Descriptive statistics is used to summarize and describe the data we already have.

Example from dataset:

- Mean (Average) height = 160.8 cm
- Minimum height = 150 cm
- Maximum height = 170 cm

👉 It answers:
"What is happening in this dataset?"

2️⃣ Inferential Statistics

Definition:
Inferential statistics is used to make predictions or generalizations about a larger population based on sample data.

Example:

- We used only 10 students (sample)
- We estimate for all students in the organization

👉 Estimated average height range (inference):
150.8 cm to 170.8 cm

🎯 Step 6: Final Understanding

- Calculating average → **Descriptive Statistics**
- Estimating range for all students → **Inferential Statistics**

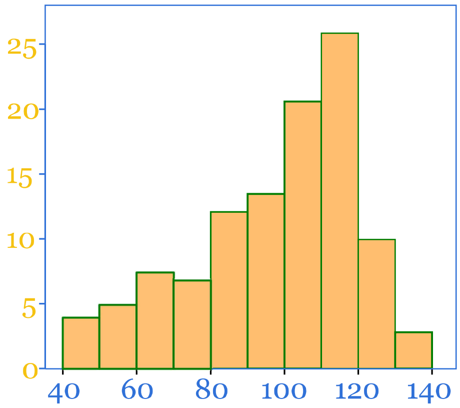
📊 Histogram and Data Distribution

📌 What is a Histogram?

A **histogram** is a graphical representation used to show the distribution of numerical data.

- The **X-axis** represents data intervals (bins)
- The **Y-axis** represents frequency (number of observations)

📈 Histogram Example



🔔 Normal Distribution

rightskew

Definition

A **normal distribution** is a symmetrical distribution where:

- Mean = Median = Mode
- Most values are concentrated in the center
- The graph forms a bell-shaped curve

➡️ Right Skewed Distribution (Positive Skew)

rightskew

Definition

A **right-skewed distribution** is a distribution where:

- The tail is longer on the right side
- Most values are concentrated on the left
- Few large values increase the average

⬅️ Left Skewed Distribution (Negative Skew)

Left

Definition

A **left-skewed distribution** is a distribution where:

- The tail is longer on the left side
- Most values are concentrated on the right
- Few small values reduce the average

📊 Summary Table

Distribution Type	Shape	Tail Direction	Characteristics
Normal	Bell-shaped	No skew	Mean = Median = Mode
Right Skewed	Asymmetrical	Right	Few large values
Left Skewed	Asymmetrical	Left	Few small values

📊 Variance & Standard Deviation (Both Datasets)

📌 Dataset 1: 69, 70, 71

✅ Step 1: Mean

$\bar{x} = (69 + 70 + 71) / 3$ $\bar{x} = 210 / 3$ $\bar{x} = 70$

✅ Step 2: Deviations

Value (x)	x - 70	$(x - 70)^2$
69	-1	1
70	0	0
71	1	1

✅ Step 3: Variance

$\sigma^2 = (1 + 0 + 1) / 3$ $\sigma^2 = 2 / 3$ $\sigma^2 = 0.67$

✅ Step 4: Standard Deviation

$\sigma = \sqrt{0.67}$ $\sigma \approx 0.82$

✅ Final Answer (Dataset 1)

- Variance = **0.67**
- Standard Deviation = **0.82**

📌 Dataset 2: 50, 70, 90

✅ Step 1: Mean

$\bar{x} = (50 + 70 + 90) / 3$ $\bar{x} = 210 / 3$ $\bar{x} = 70$

✅ Step 2: Deviations

Value (x)	x - 70	$(x - 70)^2$
50	-20	400
70	0	0
90	20	400

✅ Step 3: Variance

$\sigma^2 = (400 + 0 + 400) / 3$ $\sigma^2 = 800 / 3$ $\sigma^2 = 266.67$

✅ Step 4: Standard Deviation

σ = √266.67 σ ≈ 16.33

✅ Final Answer (Dataset 2)

- Variance = **266.67**
- Standard Deviation = **16.33**

💻 R Code (Both Datasets)

```
# Dataset 1
data1 <- c(69, 70, 71)

mean1 <- mean(data1)
var1 <- mean((data1 - mean1)^2)
sd1 <- sqrt(var1)

# Dataset 2
data2 <- c(50, 70, 90)

mean2 <- mean(data2)
var2 <- mean((data2 - mean2)^2)
sd2 <- sqrt(var2)

# Output
mean1; var1; sd1
mean2; var2; sd2
```

📌 Expected Output

Dataset 1 → Mean = 70, Variance = 0.67, SD = 0.82 Dataset 2 → Mean = 70, Variance = 266.67, SD = 16.33

In [ ]: